Barbara Draghi

Machine Learning Specialist | PhD Candidate

barbara.draghi01@gmail.com +44 (0) 7493845432 www.linkedin.com/in/barbara-draghi

DoB: 10/06/1995 **Nationality**: Italian, Eligible to work in the UK.

Summary

Machine Learning Specialist with 5 years' experience building and scaling synthetic data solutions for healthcare research and public sector innovation. I led the design and delivery of CPRD's Emulate Synthetic Data Platform, translating cutting-edge research into robust, client-facing capability. My research focuses on detecting and mitigating bias in synthetic health data, with peer-reviewed publications and international conference contributions.

Professional Experience

Dec 2020–Present

Machine Learning Specialist, *Clinical Practice Research Datalink – London, United Kingdom*

- Five years of experience contributing to CPRD's Synthetic Data Project, designing, developing, and deploying solutions for generating high-fidelity synthetic primary care data.
- Led the technical development of CPRD Emulate Synthetic Data Platform, transforming it from a proof-of-concept into a robust self-service system supporting external client use.
- Delivered the first end-to-end proof-of-concept for a synthetic data platform to demonstrate feasibility, leading to stakeholder engagement and future investment.
- Contributed to synthesising real-world primary healthcare data, utilising machine learning (ML) approaches. My contributions focused on optimising existing pipelines and developing new frameworks to enhance the utility and reliability of synthetic data. In particular, I developed scalable pipelines for synthetic CPRD COVID-19 data generation, optimised using SparkR in Databricks, reducing runtime from over 1 hour to ~30 minutes and simplifying execution for non-technical users.
- Contributed to building a synthetic relational database using Bayesian approaches as part of a commissioned service for the Berlin Center for Epidemiology and Health Research.
- Developed a comprehensive framework for analysing, evaluating, and comparing synthetic data to RWD, addressing the need for standardised assessment methods. This framework is now used used to evaluate synthetic data for various commissioned services, providing reliable data quality assessments and fostering trust among data owners and stakeholders.
- Designed and deployed machine learning models to identify and mitigate bias in primary care EHR data, contributing to award-nominated research and peer-reviewed publications.
- Contributed to led CPRD's Data Science Mentorship Programme, supporting cross-team technical development, creating training content, and promoting an inclusive learning environment.

Technologies: Docker, Python, R, SQL, Visual Studio, Agile, Azure, Visual Studio Code, GIT.

Aug 2020 – Nov 2020

Data Scientist Researcher, *University of Pavia – Pavia, Italy, Aug 2020 – Nov 2020*

- **Data analysis** for the identification of cancer patients' needs using R (CAPABLE EU project)

Technologies: R, Microsoft Excel

Sep 2019 – Mar 2020

Computer Science Student Intern, *Brunel University - London, United Kingdom*

- Selected as one of 91 participants for a 6-months research traineeship.
- Developed a new **machine learning** algorithm for the identification of temporal disease phenotypes in patients affected by diabetes.

Technologies: R

Education

2023 – Present

Part-time PhD in Computer Science, *Brunel University London*

- Developing machine learning methods to detect and reduce bias in synthetic health data, supporting fairness and equity in healthcare research.
- Contributing to CPRD's synthetic data strategy and promoting its visibility through publications and international conference presentations.

2017-2020

Master's degree in biomedical computer engineering, *University of Pavia,*
Final grade: 110/110 cum Laude

- Relevant coursework: Medical Equipment, Software Engineering, Data Engineering, Artificial Intelligence, Object-oriented concurrent programming.

2014-2017

Bachelor's degree in biomedical computer engineering, *University of Pavia*
Final grade: 110/110 cum Laude
Relevant coursework: Procedural Programming, Algorithms, Databases, Web Development

Languages

Italian (native)
English (advanced)
German (basic)

Research, Awards & Engagement

Published Machine Learning research works in peer-reviewed scientific journals and International Conferences:

- B. Draghi, Z. Wang, P. Myles, and A. Tucker, 'Identifying and handling data bias within primary healthcare data using synthetic data generators', Heliyon, p. e24164, 2024, doi: https://doi.org/10.1016/j.heliyon.2024.e24164.
- Draghi, B., Wang, Z., Myles, P. & Tucker, A. (2021). BayesBoost: Identifying and Handling Bias Using Synthetic Data Generators. In Proceedings of the Third International Workshop on Learning with Imbalanced Domains: Theory and Applications, in Proceedings of Machine Learning Research. Available from https://proceedings.mlr.press/v154/draghi21a.html.
- Z. Wang, B. Draghi, Y. Rotalinti, D. Lunn, and P. Myles, 'High-Fidelity Synthetic Data Applications for Data Augmentation', Artificial Intelligence. IntechOpen, Jan. 12, 2024. doi: 10.5772/intechopen.113884.
- Khoshravan Azar, A., Draghi, B., Rotalinti, Y., Myles, P., Tucker, A. (2023). The Impact of Bias on Drift Detection in AI Health Software. In: Juarez, J.M., Marcos, M., Stiglic, G., Tucker, A. (eds) Artificial Intelligence in Medicine. AIME 2023. Lecture Notes in Computer Science(), vol 13897. Springer, Cham. https://doi.org/10.1007/978-3-031-34344-5_37

Activities and Achievements:

- **Winner** of the **best presentation award** during the CSBPS 2023: Computer Science Brunel PhD Symposium 2023, Brunel University London, UK, April 26, 2023
- Participated as a **panelist** at the HL7 FHIR Synthetic Data Event, an international event where I gave my perspective on Fairness in Synthetic Data.
- **Nominated** for the Public Sector Awards for **Excellence in Central Government** thanks to the important research work I carried out to create a machine learning tool for data bias detection and correction in primary healthcare data in the MHRA.

Certifications

2020 Professional qualification for **Biomedical Engineering** passed in Pavia, Italy.
2023 **Python**, issued by HackerRank
2024 **SQL**, issued by HackerRank

barbara.draghi01@gmail.com +44 (0) 7493845432